

Game Theory and Auctions

Business Economics, Climate Change and the Environment

Simon Quinn

Module outline

LECTURE	DATE	TOPIC
1	9 September	Growth, Transformation and Transition
2	11 September	Introduction to Game Theory
3	16 September	Game Theory and Auctions
4	17 September	Game Theory and the Environment
5	23 September	Academic Papers Explained
6	25 September	Academic Papers Explained
7	30 September	Consumer Preferences
8	2 October	Institutions, Power and Inequality
9	7 October	Supply and Demand
10	9 October	Market Successes and Failures

A question on 'Academic Papers Explained'

Question: What are we optimizing for in this project? Is it more focused on deeply analysing the content, limitations, and methodology of this paper? Or is it more focused on teamwork and how well we can present a summary of business/climate topics to our classmates?

A question on 'Academic Papers Explained'

Question: What are we optimizing for in this project? Is it more focused on deeply analysing the content, limitations, and methodology of this paper? Or is it more focused on teamwork and how well we can present a summary of business/climate topics to our classmates?

Answer: Both objectives are important -- but certainly you should work on analysing the content, limitations and methodology of the paper.

A question on the clickable links on slides

Question: If only a certain graph from your slides is part of the linked textbook chapter, for the exam are we only required to know the content related to the graph on the slides, or should we know everything in that chapter, even if the content / graphs aren't on your slides?

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Answer: The reading list is on the 'Files' tab ('Module Outline'), and you should follow this -- the clickable links are just a matter of convenience.

A question on the slides for Academic Papers Explained

Question: When are the slides due?

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Answer: Please upload these to Insendi by 5pm on 22 September (i.e. the Lecture 1 slides are correct).

Auctions and the environment

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- ▶ The European Union's Emissions Trading System;
- ▶ Procurement of renewable energy in over 100 countries;
- ▶ Buying and selling 'green hydrogen' (and derivatives), through the H2Global instrument;

...and used in many other **market scenarios**.

Stone Town, Zanzibar



London, United Kingdom

Wind power

Jillian Ambrose *Energy correspondent*

Fri 8 Sep 2023 17.28 BST



Explainer

What went wrong at UK government's offshore wind auction?

Industry sources say auction was 'catastrophic', after failure to secure any new offshore windfarms

● [UK auction secures no offshore windfarms](#)



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In the next three hours together, we will aim to:

1. Understand different kinds of auction **mechanism**;
2. Understand **intuitively** how firms should (and should not!) bid in different kinds of auctions;
3. Think about how policy should choose between different kinds of auctions.

The basic trade-off of bidding in an auction

<https://www.youtube.com/watch?v=DQ4mraAx23I>

Form 10 groups please!

Logistics check. . .

1. Learn (and remember) your group letter!
2. Choose your group representative (who will simply follow the group's instructions).
3. **Group representatives need to e-mail me now, including your group letter:**
`simon.quinn@imperial.ac.uk`.
4. Figure out how you will pay me, and how I will pay you.

Outline of auctions

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No communication with other teams!!

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6. **All-pay** (not for money): I will read from your online submission. Each team pays what it bid, and the team that bids the higher wins.

Over to you!

Submit your data please!

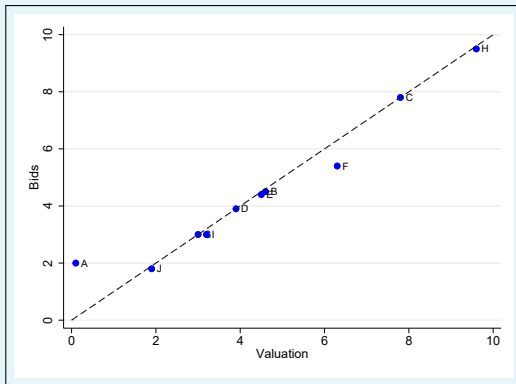
Time to play...!

	A	B	C	D	E	F	G	H	I	J
ALL-PAY										
OPEN BID	✓					✓				
ASCENDING CLOCK		✓					✓			
SECOND-PRICE SEALED BID			✓					✓		
FIRST-PRICE SEALED BID				✓					✓	
DESCENDING CLOCK					✓					✓

Some questions for discussion during the break...

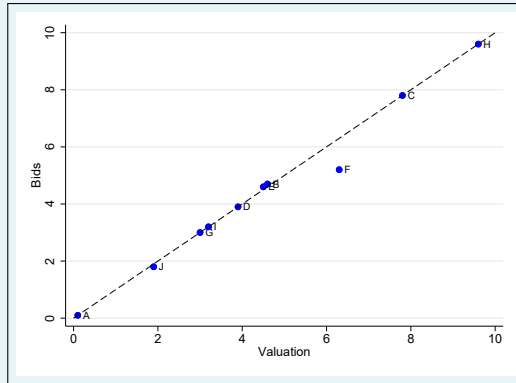
1. Which auction was **easiest/simplest** to bid in?
2. Which auction was **fairest**?

Bidding: Open bid



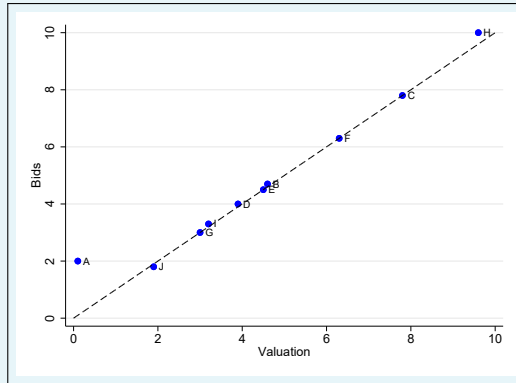
(Note: The dotted line shows a strategy of 'bid your valuation'.)

Bidding: **Ascending clock**



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Bidding: **Second-price sealed bid**



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Remember '**obviously dominant strategies**'?

Definition: Obviously Dominant strategy

For Player 1 (for example), action a is **obviously dominant** if, at the time when the player must choose, the best possible outcome from any other action is no better than the worst possible outcome from action a .

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Key concept: '**Bid your valuation**'

'Bid your valuation' is a **dominant** strategy for both the ascending clock and the second-price sealed bid auction. But it is an **obviously dominant** strategy only for the ascending clock auction.

Reasoning about the second-price sealed bid auction ...

Suppose that your group's valuation was £5...

"I want to bid £6...

- ▶ The **best outcome** from bidding £6 is that the other player bids £0 – so I make a profit of £5.
- ▶ The **worst outcome** from bidding £5 is that the other player bids above £5 – so I lose the auction and make a profit of £0.

I could do better by bidding £6 than by bidding £5."

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Bidding your valuation is **not** an obviously dominant strategy. (Note the **failure of contingent reasoning**...)

Reasoning about the ascending clock auction ...

Suppose that your group's valuation was £5, and that the auction clock has now reached £5...

"I want to stay in the auction until the price reaches bid £6...

- ▶ The **best outcome** from staying in the auction is that the other player stops the clock before £6 – so I lose the auction and make a profit of £0.
- ▶ The **worst outcome** (indeed, the only outcome) from stopping the clock at £5 is that I win the auction, pay £5, and make a profit of £0.

I **cannot possibly do better** by staying in the auction.

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Bidding your valuation is an **obviously dominant strategy**.

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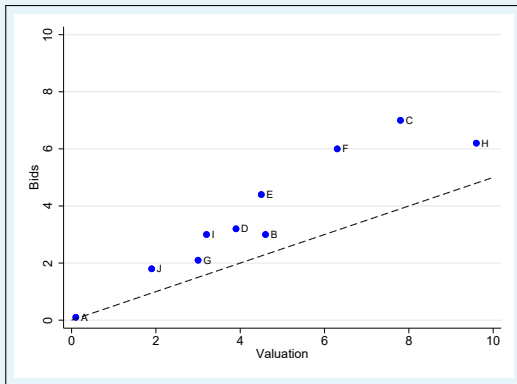
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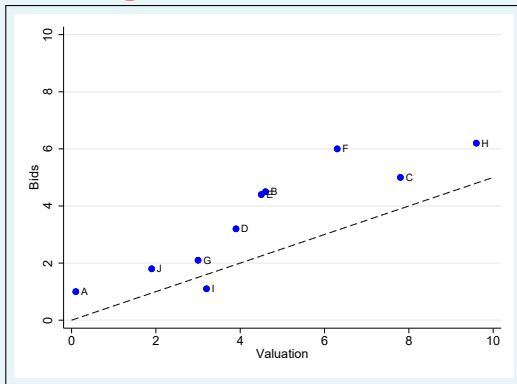
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Bidding: **First-price sealed bid**



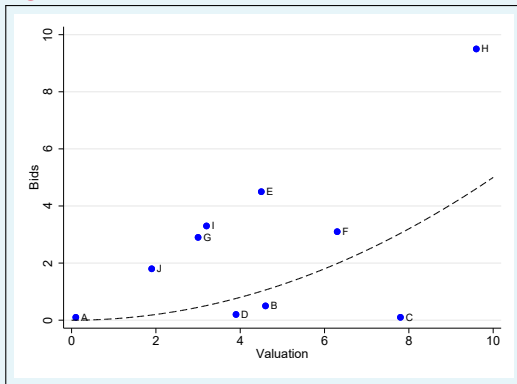
(Note: The dotted line shows a strategy of 'bid **half** your valuation'.)

Bidding: Descending clock



(Note: The dotted line shows a strategy of 'bid **half** your valuation'.)

Bidding: **All-pay**



(Note: The dotted line shows a strategy of 'bid the square of your valuation divided by 20'.)

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Note: In almost all jurisdictions, collusion in auctions is **illegal!**



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Mercury bid on several licenses and sometimes submitted bids that ended with three-digit numerical codes. Mercury used the coded bids to communicate with rival bidders during the auction. By using the codes, Mercury created links between two or more specific markets and thereby proposed to competing bidders that they not bid against each other in the linked markets.

A hypothetical example...



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I think it's the winner's **dilemma** more than a **curse**: any time you're the buyer, you wonder why the other [bidders] didn't want it.

Bruce Wasserstein (2007)

Some insights...

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3. In some auctions, your submitted bid determines **whether** you win and – conditional upon winning – **how much you win**. In this case, you should ‘**shade your bid**’.

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4. When you decide on your valuation, remember to consider the **Winner’s Curse**.

Game Theory and Auctions

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